

INCIDENT REPORT — INCIDENT-1: worker failure → queue backlog

The deepest-documented incident (PDF §19: at least one in detail). INCIDENT-2 (EHR timeout) is fully reported in `INCIDENTS.md`; drill definitions F1–F7 live there too.

1. Failure

Worker container stopped (`scripts/fail-worker.ps1`) during steady intake; 3 probe jobs buffered, then `load-test.ps1 -Count 100` burst (~103 queued).

2. Impact

Processing rate 0; depth ~103; `oldest_age_s` climbing. Zero job loss (Redis buffers). API stays green — intake and processing fail independently.

3. Detection

`WorkerDown` (up{job="worker"}==0, 2m) then `QueueBacklog` (depth>20, 5m); `/worker/status` → `alive:false`, heartbeat aging; `queue/stats` rising. Command: `scripts/alerts.ps1`.

4. Investigation

`LLEN jobs:queue=103` vs `processed` frozen; `docker compose ps worker` = exited; logs end cleanly (gone, not wedged); `docker stats` shows api idle → worker-side capacity problem. Correlation ids join `api→queue→worker→DB` rows.

5. Root cause

Worker process gone (simulated crash); single default replica = zero drain capacity until restart. Queue design (BRPOPLPUSH) held: nothing lost.

6. Evidence

[PENDING-ENGINE: `alerts.ps1` FIRING lines; `queue/stats` series; worker log tail; `worker_restarts_total` before/after; Grafana queue tile.]

7. Recovery

`scripts/recover-worker.ps1` (start, poll depth→0) → `--scale worker=3` (3× drain) → scale back to 1.

8. Verification

Depth 0, all 103 `completed`, heartbeat fresh, restarts +1, probe job attempts=1, alerts resolve.

9. Prevention

Heartbeat/depth/age alerts exist; runbook scales workers (never api) on backlog; prod-like default of 2 workers recommended.

Timeline: T+0 stop+buffer → T+2 WorkerDown → T+3 burst → T+7 QueueBacklog → T+8 restart → T+10 scale-3 → T+12 drained, resolved.